Vitaran Edutech

HINTS & SOLUTIONS

INTEGRATED PROGRAM

Biology

Class – 6

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1 FOOD -WHERE DOES IT COME FROM

Solutions

Worksheet – 1

1. **Correct Answer (C) nutrients.**

   Nutrients are components in food that provide energy and help the body grow, repair, and stay healthy. Examples: carbohydrates, proteins, fats, vitamins, minerals.

2. **Correct Answer (C) Tamil Nadu.**

   Dosa (fermented rice pancake) and rasam (spicy soup) are traditional South Indian dishes, mainly from Tamil Nadu.

3. **Correct Answer (C) Ingredients.**

   Ingredients are the raw materials used in cooking, such as rice, vegetables, spices, and oil.

4. **Correct Answer (B) Wheat flour**

   Chapati is an unleavened flatbread made from wheat flour and water.

5. **Correct Answer (A) Green gram**

   Pulses are seeds of leguminous plants used as food. Examples: green gram, chickpeas, lentils. Maize, potato, banana are not pulses.

6. **Correct Answer (C) Mango**

   Botanically, a fruit develops from the flower and contains seeds. Mango is a fruit, while onion, cabbage, and okra are vegetables (though okra is technically a fruit in botanical terms, it is commonly eaten as a vegetable).

7. **Correct Answer (D) starch**

   Plants store extra energy as starch, which is a carbohydrate. Examples: potatoes, rice, wheat.

8. **Correct Answer (C) vegetable oil**

   Seeds like sunflower, groundnut, and sesame are pressed to extract oil used for cooking.

9. **Correct Answer (C) leaves of tea plant**

   Tea leaves are harvested, dried, and processed to make tea. Roots, stems, and flowers are not used for tea.

10. **Correct Answer (B) Wheat**

    Wheat seeds (grains) are edible and used to make flour for bread, chapati, and other foods. Potato is a tuber, ginger is a rhizome, and spinach is a leafy vegetable.

WORKSHEET-2

1. **Correct Answer (B) Honey**

   Honey is produced by bees, an animal. Rice, wheat, and sugar come from plants.

2. **Correct Answer (C) Meat**

   Meat refers to the edible muscle tissue of animals. Milk, honey, and butter are animal products but not flesh.

3. **Correct Answer (C) Pisciculture**

   Pisciculture is fish farming. Apiculture = bee-keeping, Agriculture = crop cultivation, Horticulture = cultivation of fruits, vegetables, and flowers.

4. **Correct Answer (C) depend on plants for food**

   Animals cannot make their own food (unlike plants) and rely on other organisms (plants or animals) for nutrition.

5. **Correct Answer (C) Herbivores**
   Herbivores eat only plants (e.g., cow, deer). Carnivores eat meat, omnivores eat both plants and animals, scavengers feed on dead animals.

6. **Correct Answer (C) Carnivores**
   Carnivores like lions and tigers feed on meat.

7. **Correct Answer (C) parasites**
Parasites live on or in a host and obtain food at the host's expense.
8. **Correct Answer (B) Ectoparasites**
Ectoparasites live on the outside of the host (e.g., lice, ticks).Endoparasites live inside the host.
9. **Correct Answer (C) Omnivores**
Humans eat both plants and animals, so they are omnivores
10. **Correct Answer (B) Earthworm**
Decomposers break down dead organic matter into simpler substances. Earthworms decompose soil matter, whereas vulture is a scavenger, lion is a carnivore, sparrow is a bird/consumer.

WORKSHEET-3

1. **Correct Answer (C) Food chain**
A food chain shows a linear sequence of organisms where each one feeds on the previous one.
2. **Correct Answer (B) Who eats whom**
It shows the feeding relationship between organisms in an ecosystem.
3. **Correct Answer (C) Different food chains**
Food chains vary depending on the plants and animals present in that particular environment.
4. **Correct Answer (C) Food web**
A food web is a network of interconnected food chains showing multiple feeding relationships.
5. **Correct Answer (C) Food web is formed by many food chains**
Unlike a single food chain, a food web represents complex interconnections of multiple food chains.
6. **Correct Answer (C) Interdependence**
Each organism depends on others for food, showing interdependence in an ecosystem.
7. **Correct Answer (B) Surroundings**
Animals gather food from their environment to survive.
8. **Correct Answer (B) Natural environments**
Food chains naturally exist in forests, oceans, and grasslands. Artificial setups like labs or aquariums may mimic them but are not natural.
9. **Correct Answer (C) Web-like network**
Because multiple food chains intersect, forming a network rather than a single straight line.
10. **Correct Answer (A) Energy flow**
Energy flows from producers → primary consumers → secondary consumers → tertiary consumers, illustrating how energy moves in nature.

COMPETITIVE CORNER

SINGLE OPTION CORRECT TYPE

1. **Correct Answer (D) Dosa**
Dosa is a traditional South Indian dish, especially popular in Tamil Nadu. Dal-bati → Bihar, Litti-chokha → Bihar, Sattu → Bihar/UP.
2. **Correct Answer (C) Salt**
Salt is a mineral, not derived from plants or animals. Eggs → animal, Chilies & Onions → plant.
3. **Correct Answer (B) Honey**
Bees collect nectar from flowers and convert it into honey. Yogurt → milk product, Ghee → clarified butter, Sattu → ground pulses.
4. **Correct Answer (C) Bear**
Omnivores eat both plants and animals. Bears eat berries, fish,insects, and small animals. Lion → carnivore, Cow → herbivore, Snake → carnivore.
5. **Correct Answer (C) Seawater**
Most edible salt is obtained by evaporating seawater. It can also come from salt mines (underground rocks), but seawater is the major natural source.
6. **Correct Answer (A) Grass**
Producers are organisms that make their own food using sunlight

(photosynthesis). Grass → producer, others → consumers.

7. **Correct Answer (C) Plants (Producers)**
Food chains begin with producers (plants) that create energy from sunlight, followed by consumers.
8. **Correct Answer (B) Flower**
Broccoli and cauliflower are clusters of flower buds. Root → carrot,Stem → sugarcane, Seed → wheat.
9. **Correct Answer (B) Because they feed on and break down dead plants and animals**
Decomposers recycle nutrients by breaking down dead organic matter.
10. **Correct Answer (C) Flavouring, colouring, or preserving food**
Spices like cardamom, cloves, cumin, and saffron are added mainly for taste, aroma, color, and preservation, not for mass or water content.

ONE OR MORE THAN ONE OPTION CORRECT TYPE

11. **Correct Answer (A, B, C)**

A) Supplying energy for daily activities

B) Repairing damaged tissues and healing wounds

C) Protecting the body from diseases by strengthening the immune System.

Energy giving – Carbohydrates and fats provide energy for movement, thinking, and daily work.

Body building and repair – Proteins help in growth, repair of damaged tissues, and healing wounds.

Protection and regulation – Vitamins and minerals strengthen the immune system and protect against diseases.

12. **Correct Answer (A, C, D)**
Onion – A vegetable grown from plants.
Chillies – Fruits of chilli plants.
Spices – Most spices (like pepper, turmeric, cardamom, cloves) come from different parts of plants such as seeds, bark, roots, or fruits.
Salt – Obtained from natural sources like seawater and mineral rocks.
13. **Correct Answer (A, B, C)**
Paneer – Made by curdling milk using lemon juice or vinegar.
Butter – Made by churning cream obtained from milk.
Ghee – Clarified butter prepared by heating butter to remove water content.
Rice – A plant product (cereal grain), not derived from milk.
14. **Correct Answer (A, B, D)**
Animals are called consumers because they cannot make their own food.
They depend on plants (producers) either directly (herbivores) or
Indirectly (carnivores and omnivores).
They consume other organisms to obtain energy.
15. **Correct Answer (A, C)**
A food chain shows the sequence of organisms where each one is eaten by the next.
It represents who eats whom and how energy flows from:
Producer → Herbivore → Carnivore → Top consumer
Example:
Grass → Deer → Tiger

ASSERTION AND REASON TYPE

16. **Correct Answer (A)**

Both Assertion (A) and Reason (R) are true and Reason (R) is the correct explanation of Assertion (A).

Green plants are called producers because they make their own food through photosynthesis using sunlight, carbon dioxide, and water. This ability to produce food is the reason they are placed at the base of the food chain.

17. **Correct Answer (A)**

Both Assertion (A) and Reason (R) are true and Reason (R) is the correct explanation of Assertion (A).

A food web is more stable because it has many connected food chains.

If one animal disappears, others can still find alternative food sources.

18. Correct Answer (A)

Both Assertion (A) and Reason (R) are true and Reason (R) is the correct explanation of Assertion (A).

Tea is made from the leaves of the tea plant because the leaves contain special natural chemicals that give tea its taste and aroma.

19. Correct Answer (A)

Both Assertion (A) and Reason (R) are true and Reason (R) is the correct explanation of Assertion (A).

Rice and wheat are the seeds of grass plants, so they are called cereals.
The reason correctly explains why rice and wheat are classified as cereals.

MATRIX MATCHING TYPE

20. Correct Answer (A)

(A) a→Q, b→P, c→R, d→S

21. Correct Answer (A)

(A) a→Q, b→R, c→S, d→P

COMPREHENSION TYPE

PASSAGE-1

22. Correct Answer (B) kidney beans

23. Correct Answer (B) Seeds

24. Correct Answer (B) Pulses

PASSAGE-2

25. Correct Answer(D) Sugar

26. Correct Answer (B) Spices

27. Correct Answer (B) Sugarcane and sugar beet

PREVIOUS YEAR QUESTIONS

1. Correct Answer (B) M N O V I R O E S

When you unscramble the letters in Option B, they form the word OMNIVORES, which is the term used for animals that eat both plants and the flesh of other animals.

2. Correct Answer (A) Sea food, Meat

L (Sea food): Marine environments are naturally high in iodine. Fish, shellfish, and seaweed absorb this mineral from seawater, making them some of the richest natural sources available.

M (Meat): While phosphorus is found in many foods, meat and animal products (alongside dairy) are highly concentrated sources of phosphorus, which is essential for bone health and cellular repair.

3. Correct Answer (C) Fish, Papaya, Rice

Why Option C is wrong: While Fish is a great protein source, Papaya is primarily a source of vitamins (A and C), and Rice is a source of carbohydrates (energy-giving food). They do not belong in a group exclusively for body-building foods.

4. Correct Answer (D)

To find the right group, we categorize each animal based on its natural diet:
P (Herbivores) – Eat only plants:
(vi) Cow
(viii) Goat
Q (Carnivores) – Eat only other animals:
(ii) Lion
(iii) Tiger
(iv) Spider (eats insects)
(v) Lizard (eats insects)
R (Omnivores) – Eat both plants and animals:
(i) Crow (eats grains and small animals/carrion)
(vii) Human being
(ix) Bear (eats berries/honey and fish/meat)

2 Components of food

Solutions

Worksheet – 1

1. **Correct answer – C**

   Adding iodine solution to a food sample confirms the presence of starch if it turns blue-black.

2. **Correct answer – B**

   The text states that excess body fat leads to a condition called obesity, which can be harmful and lead to heart problems.

3. **Correct answer – C**
   Consuming a variety of foods ensures the body receives all the chemicals it needs for health.
4. **Correct answer – A**
   Water is a vital component of food that must be consumed in sufficient amounts to keep the body healthy.
5. **Correct answer – C**
   Carbohydrates are found in our diet primarily in the forms of complex starch and various sugars.
6. **Correct answer – A**
   Every gram of carbohydrates provides the body with four kilocalories of energy.
7. **Correct answer – C**
   Proteins are complex chemical substances built from smaller units called amino acids.
8. **Correct answer – A**

   Proteins only serve as an energy source once the body's primary energy stores are used up.

9. **Correct answer – B**

   The primary role of fats is to serve as a reserved form of energy stored within the body.

10. **Correct answer – C**

    The egg yolk is a source of fat which is yellow part of the egg.

worksheet – 2

1. **Correct answer – C**
   Retinol is the specific chemical name provided for Vitamin A.
2. **Correct answer – B**
   Approximately 4% of total body weight is made up of minerals.
3. **Correct answer – C**
   Dehydration is the term used to describe the state when the body loses an excessive amount of water.
4. **Correct answer – A**
   Fluorine is the mineral specifically responsible for creating the enamel on teeth.
5. **Correct answer – B**
   Common salt is often fortified with iodine to make iodized salt, which provides the mineral iodine needed for proper growth and thyroid function
6. **Correct answer – B**
   Fat-soluble vitamins are absorbed with fats and stored within the body's fat tissues and liver.
7. **Correct answer – D**
   Iodine is essential for ensuring the nervous system works correctly.
8. **Correct answer – C**
   Water is a major component of the human body, accounting for about 70% of its weight.
9. **Correct answer – C**
   Though essential for digestion, dietary fibre does not provide nutrients to the body.
10. **Correct answer – B**

    Iron is required for the formation of Haemoglobin, the red pigment that carries oxygen in the blood.

worksheet 3

1. **Correct answer – B**

   Balance diet contains all the essential nutrients require to the body in right proportion.

2. **Correct answer – C**

   all these symptoms are for Kwashiorkor

3. **Correct answer – B**

   the diet is deficient in both protein and carbohydrate, it causes marasmus

4. **Correct answer – B**

   Beriberi, caused by Vitamin B1 deficiency, specifically results in extreme weakness and a sour mouth.

5. **Correct answer – C**

   Iodine is essential for the thyroid gland to function, and it's lack causes the gland to swell, a condition known as Goitre.

6. **Correct answer – B**

   Water-soluble vitamins and minerals present on the surface of grains are washed away with repeated rinsing.

7. **Correct answer – A**
   Both Vitamin B12 and Iron are necessary for healthy blood, and their deficiency leads to Anaemia and low energy.
8. **Correct answer – B**
   Vitamin A is critical for eye health, and its deficiency makes it difficult to see in low-light conditions.
9. **Correct answer – B**

   The skins of many fruits and vegetables contain a high concentration of nutrients that are lost when peeled away.

10. **Correct answer – B**

    Oxygen reacts with the cut surfaces of fruit, causing oxidation which destroys sensitive vitamins and alters the appearance.

Competitive Corner

SINGLE OPTION CORRECT TYPE

1. **Correct answer – B**

   Scurvy is caused by a Vitamin C deficiency, which is essential for collagen production to heal wounds and keep gums healthy.

2. **Correct answer – A**

   The Biuret test (using copper sulphate and sodium hydroxide) turns a food sample violet if proteins are present.

3. **Correct answer – C**

   While roughage is important for digestion, vitamins and minerals are specifically classified as the remaining two major nutrient groups.

4. **Correct answer – C**

   A balanced diet is defined as one that contains all necessary nutrients, including fiber (roughage) and water, in the correct proportions.

5. **Correct answer – A**
   Roughage, or dietary fiber, adds bulk to the diet and helps the body move undigested food through the digestive system.
6. **Correct answer – C**
   Minerals are considered micronutrients because the body only needs them in small quantities to function properly.
7. **Correct answer – D**

   According to your text, deficiency diseases occur when essential nutrients are missing from the diet over a long period.

8. **Correct answer – C**

   Calcium, phosphorus, iron, and iodine are all inorganic elements classified as minerals in nutrition.

9. **Correct answer – D**

   While meat contains iron and zinc, fruits like bananas are specifically recognized in many school curriculum contexts as rich sources of minerals like potassium.

10. **Correct answer – B**

    Macronutrients are nutrients required in large amounts (Carbohydrates, Fats, and Proteins), whereas vitamins and minerals are micronutrients.

ONE OR MORE THAN ONE OPTION CORRECT TYPE

11. **Correct answer – A, B, C, D**
12. **Correct answer – A, B**
    Rice is a major source of complex carbohydrates (starch), while Mango is a rich source of simple carbohydrates (sugar); meat, however, is primarily a protein source.
13. **Correct answer – A, D**
    Micronutrients are substances like vitamins and minerals that the body requires in small amounts to function properly, unlike macronutrients (fats, proteins, and carbs) which are needed in larger quantities.
14. **Correct answer – A, B**

    Vitamins B and C are both water-soluble, while Vitamins A, D, E, and K are fat-soluble
15. **Correct answer – B, C, D**

    Carbohydrates are organic compounds specifically made up of Carbon, Hydrogen, and Oxygen atoms.

ASSERTION AND REASON TYPE

16. **Correct answer – A**
    The reason correctly explains why the practice is discouraged to preserve nutritive value.
17. **Correct answer – B**
    Both are true, but the reason describes the symptoms of the disease rather than explaining why iron deficiency causes it (which would involve hemoglobin/blood production).
18. **Correct answer – D**

    The assertion is false because "unlimited" consumption is harmful; the reason is true as it highlights the risks of excess fat.
19. **Correct answer – C**

    The assertion is true (they protect from disease), but the reason is false because they are actually required in **small** quantities.

MATRIX MATCHING TYPE

20. **Correct answer – B**

    a-r, b-s, c-p, d-q, e-t
21. **Correct answer – D**

    a-r, b-p, c-s, d-t, e-q

COMPREHENSION TYPE

22. **Correct answer**

    1. B

    2. C

    3. A
23. **Correct answer**

    1. B

    2. D

    3. C

Previous Year Questions

1. **Correct answer – C**

   Goitre is caused by a deficiency of Iodine, whereas Vitamin A deficiency leads to night blindness.
2. **Correct answer – D**

   Shruti's meal is balanced because it contains all essential components: carbohydrates (chapati), proteins (pulses), fats (ghee), and vitamins/minerals (vegetables/fruit).
3. **Correct answer – B**

   Sodium (X) is essential for nerve function, Iron (Y) is needed for Haemoglobin, and Calcium (Z) maintains bone and tooth strength.
4. **Correct answer – D**

   The purple color change indicates a positive Biuret test, which confirms the presence of protein found in eggs.

5. Correct answer – B

Statement (ii) is incorrect because Kwashiorkor is specifically caused by a severe deficiency of protein, not carbohydrates or fats.

6. Correct answer – A

Extra carbohydrates are stored in the body as fat, which can lead to obesity and other health issues if consumed in excess.

7. Correct answer – D

All the statements (A, B, and C) provided are scientifically accurate descriptions of nutrient deficiencies and their effects.

8. Correct answer – B

Anaemia results from a lack of Haemoglobin, the substance in red blood cells that carries oxygen to the body's tissues.

9. Correct answer – C

While fish is a body-building protein, rice is an energy-giving carbohydrate and papaya is a protective food rich in vitamins.

10. Correct answer – A

Vitamin A deficiency causes night blindness (W), sunlight (X) provides Vitamin D, Vitamin K (Y) prevents excessive bleeding, and Vitamin E deficiency causes nerve/muscle weakness (Z).

3 Let us understand Plants

Solutions

Worksheet – 1

1. **Correct answer – B**

   Plants provide oxygen and are the primary source of food and shelter.

2. **Correct answer – C**

   Shrubs branch near the ground; trees have a main trunk and high branches.

3. **Correct answer – D**

   Creepers grow horizontally along the soil.

4. **Correct answer – B**

   Cactus stems take over the role of leaves to save water and make food.

5. **Correct answer – C**

   Roots grip the soil, which prevents it from being washed or blown away.

6. **Correct answer – D**

   Potato, ginger, and turmeric are examples of modified underground stems.

7. **Correct answer – C**

   Spines prevent water loss in the desert and stop animals from eating the cactus.

8. **Correct answer – B**

   Roots are the "life-support mechanism" for water and mineral intake.

9. **Correct answer – B**

   Prop roots/aerial roots are designed for additional structural support.

10. **Correct answer – C**

    Wheat, rice, and onions have bushy, thread-like fibrous roots.

worksheet – 2

1. **Correct answer – C**

   Whorled phyllotaxy is defined as having three or more leaves arranged in a circle at the same node.

2. **Correct answer – A**

   A phyllode is a leaf modification where the petiole becomes flattened and green to perform photosynthesis.

3. **Correct answer – B**

   Insectivorous plants like the Pitcher plant trap insects primarily to supplement nutrients like nitrogen that are missing from the soil.

4. **Correct answer – B**

   Insectivorous plants like the Pitcher plant trap insects primarily to supplement nutrients like nitrogen that are missing from the soil.

5. **Correct answer – C**

   Chlorophyll is the green pigment that traps solar energy to drive the chemical reaction of photosynthesis.

6. **Correct answer – C**

   Glucose produced during photosynthesis is converted into starch for long-term storage in various plant parts.

7. **Correct answer – B**

   Stomata are microscopic oval-shaped openings on the leaf surface used for gas exchange and water release.

8. **Correct answer – B**

   Transpiration is the evaporation of water from the plant's surface, specifically through the stomatal openings.

9. **Correct answer – C**

   Transpiration occurs during the day when sunlight is present, temperatures are higher, and stomata are open.

10. **Correct answer – B**

    The loss of water from leaves creates a suction force (transpirational pull) that draws water and minerals up from the roots.

worksheet 3

1. **Correct answer – C**

   Sepals are the outermost green leaf-like parts that protect the flower during its bud stage.

2. **Correct answer – B**

   Bright colors and thin structures of petals are designed to catch the attention of insects to assist in reproduction.

3. **Correct answer – B**

   The stamen is the male part of the flower, consisting of the stalk-like filament and the pollen-bearing anther.

4. **Correct answer – C**

   Pollen grains are the fine, powdery substances found in the anther that are essential for the pollination process.

5. **Correct answer – C**

   Self-pollination is specifically defined as the transfer of pollen within the same flower.

6. **Correct answer – C**

   Following fertilization, the swollen base of the carpel, known as the ovary, matures and develops into the fruit.

7. **Correct answer – B**

   The ovules are the "future seeds" of the plant that transform after the fertilization process is complete.

8. **Correct answer – B**

   Dicot seeds are characterized by having two cotyledons, which also act as the first seed leaves.

9. **Correct answer – B**

   Reticulate or net-like venation is a distinct leaf characteristic of plants that grow from dicot seeds.

10. **Correct answer – C**

    Corn is a monocot because it has a single cotyledon and exhibits parallel venation in its leaves.

Competitive Corner

SINGLE OPTION CORRECT TYPE

1. **Correct answer – B**

   Transpiration is the biological process where plants release excess water in the form of vapor through tiny pores called stomata on their leaves, helping the plant stay cool and move nutrients upward.

2. **Correct answer – C**

   In a taproot system, the dominant central root grows downward and produces smaller side branches known as lateral roots.

3. **Correct answer – A**

   The petiole is the thin stalk that attaches the flat green portion of the leaf (the lamina) to the plant's stem.

4. **Correct answer – D**

   Stomata are microscopic pores on the leaf surface that allow the plant to "breathe" by exchanging oxygen and carbon dioxide with the air.

5. **Correct answer – B**

   Ginger is an underground modified stem called a rhizome; while radish, carrot, and sweet potato are all modified roots.

6. **Correct answer – C**

   Each stoma is surrounded by two guard cells which are typically bean-shaped (or kidney-shaped) and control the opening and closing of the pore.

7. **Correct answer – B**

   Grasses do not have a main central root; instead, they have a cluster of thin, thread-like roots of similar size that grow from the base of the stem.

8. **Correct answer – B**

   The root system is the primary organ responsible for drawing water and essential nutrients from the soil into the plant.

9. **Correct answer – B**

   Minerals are dissolved in the water present in the soil, which is then taken up by the roots and transported throughout the plant.

10. **Correct answer – C**

    There is a strong biological link between these two features; plants like grass or wheat have both fibrous roots and veins that run side-by-side in their leaves.

ONE OR MORE THAN ONE OPTION CORRECT TYPE

11. **Correct answer – A, B, C**

    Leaves perform photosynthesis (making food), transpiration (releasing water vapor), and gas exchange (breathing), while roots handle water absorption.

12. **Correct answer – A, B, C**

    According to the question, potato, ginger, and turmeric are all examples of modified underground stems that store food.

13. **Correct answer – A, C**

    The primary role of the stem is to carry water, nutrients, and food between the roots and the rest of the plant.

14. **Correct answer – A, B, C**

    Plants provide medicines, timber, and food, but wool is an animal product obtained from sheep or goats.

15. **Correct answer – A, B, C**

    Mint, tomato, and coriander are small plants with soft green stems, whereas Neem is classified as a tall, woody tree.

ASSERTION AND REASON TYPE

16. **Correct answer – A**
    The storage of prepared food is why starch is called "reserved."

17. **Correct answer – B**
    Both are true statements from the text, but the fact that it adds moisture to the air is not the *reason* why the plant itself cools down—the "sweating" effect is.

18. **Correct answer – D**
    Assertion is false; that is self-pollination. Reason is a true definition of pollination.

19. **Correct answer – C**
    Assertion is true, but Reason is false; ovules are found in the ovary, not the anther.

MATRIX MATCHING TYPE

20. **Correct answer – A**

    a- Q, b- S, c- T, d- R, e- P

21. **Correct answer – C**

    a- S, b- R, c- T, d- Q, e- P

COMPREHENSION TYPE

Passage1

22. **Correct answer – A**

    Photosynthesis is the process by which plant make their food.

23. **Correct answer – B**

    By the process of photosynthesis, the plant made food and stored in the form of starch.

24. **Correct answer – B**

    The tiny pores present on the surface of plant which helps in gases exchange is Stomata.

Passage2

25. **Correct answer – C**

26. **Correct answer – C**

27. **Correct answer – B**

Previous Year Questions

1. **Correct answer – B**

   Crocodiles are reptiles and breathe through lungs, not gills, even when they are underwater.

2. **Correct answer – C**

   Both pea and money plants have weak stems and use specialized structures (like tendrils) to climb up supports.

3. **Correct answer – B**

   This is incorrect because the pea plant is a climber, while the watermelon is a creeper.

4. **Correct answer – A**

   Sweet pea leaves are modified into tendrils for climbing (P), and lettuce is a leafy vegetable that stores food in its leaves (Q).

5. **Correct answer – D**

   Stamens are the male reproductive parts, and the petiole is the stalk that connects a leaf blade to the stem.

4 Living and Non-Living

Solutions

Worksheet – 1

1. **Correct answer – C**

   Insects are living things because they grow, breathe, and reproduce, whereas water, air, and buildings do not.

2. **Correct answer – B**

   Living things depend on non-living elements to carry out life processes like breathing (air) and making food (sunlight).

3. **Correct answer – A**

   Growth is defined as a permanent increase in size as an organism develops from a young stage to adulthood.

4. **Correct answer – B**

   Respiration is the process by which living beings breathe to release energy from the food they consume.

5. **Correct answer – D**

   While plants don't walk, they show movement through internal fluid changes or reacting to touch and support.

6. **Correct answer – C**

   A rock does not breathe, grow, or reproduce, meaning it lacks the essential characteristics of life.

7. **Correct answer – A**

   The term "oviparous" specifically refers to animals that reproduce by laying eggs rather than giving birth to live young.

8. **Correct answer – C**

   Death is the final stage for a living being once its biological processes, like respiration and nutrition, cease to function.

9. **Correct answer – B**

   Growth is a fundamental characteristic of life where organisms undergo a permanent increase in size and complexity as they age.

10. **Correct answer – B**

    All living organisms require non-living resources like water for hydration and air for breathing to stay alive.

worksheet – 2

1. **Correct answer – C**

   A seed is in a state of "biological sleep" or inactivity until it encounters the right environment to wake up.

2. **Correct answer – D**

   The embryo is the complete "baby" version of the plant tucked safely inside the seed's protective coat.

3. **Correct answer – B**

   The radicle emerges first because the plant needs to anchor itself and start absorbing water before it can grow upward.

4. **Correct answer – A**

   The plumule is the embryonic shoot, and its job is to reach the surface to find sunlight for photosynthesis.

5. **Correct answer – D**

   While fertilizers help plants grow later, a seed only needs water, air, and the right temperature to actually begin germinating.

6. **Correct answer – C**

   The cycle always starts with the seed, which contains the potential life that will eventually become a full plant.

7. **Correct answer – B**

The cycle always starts with the seed, which contains the potential life that will eventually become a full plant.

8. **Correct answer – B**

In the mature stage, the plant focuses on physical strength and size, expanding its root system and stems.

9. **Correct answer – C**

Pollination is the reproductive act that triggers the flower to begin developing new seeds for the next generation.

10. **Correct answer – B**

Before the cycle can start over, the new seeds must be spread to different locations to find space and resources to grow.

worksheet 3

1. **Correct answer – C**

The word "Amphibian" refers to their ability to live both in aquatic environments and on land.

2. **Correct answer – B**

This is the biological process of changing from an immature form (tadpole) to an adult form (frog).

3. **Correct answer – C**

Tadpoles are fully aquatic and rely on gills to extract oxygen from the water before they develop lungs.

4. **Correct answer – B**

Frog limbs always sprout in a specific order, starting with the back legs near the tail.

5. **Correct answer – D**

This is the transitional stage where the frog has legs but has not yet fully absorbed its tail.

6. **Correct answer – C**

Microorganisms are too small for the human eye, so a microscope is used to magnify them.

7. **Correct answer – A**

Good bacteria live in our gut to help break down food and fight off bad germs.

8. **Correct answer – B**

Microbes like bacteria and fungi are used in fermentation to turn milk into curd and help bread rise.

9. **Correct answer – C**

While many microbes are good, the specific term for those that cause illness or disease is "pathogens."

10. **Correct answer – B**

Scientists use various microorganisms to create antibiotics and other life-saving medical treatments.

Competitive Corner

SINGLE OPTION CORRECT TYPE

1. **Correct answer – C**

Non-living things lack internal life force and only move via external forces like wind or engines.

2. **Correct answer – A**

Insects are living beings because they breathe, eat, grow, and reproduce.

3. **Correct answer – B**

The radicle is the embryonic root that grows downward to provide the plant with nutrients.

4. **Correct answer – D**

Once leaves unfurl, the seedling uses sunlight to convert CO_2 and water into food.

5. **Correct answer – B**

Air (oxygen) is required for the seed's metabolism, and loose soil allows roots to move.

6. **Correct answer – A**

Most seeds germinate underground in the dark; they only need light after they become seedlings.

7. **Correct answer – C**

This jelly acts as a shock absorber and prevents the eggs from drying out.

8. **Correct answer – B**

The tadpole is the larval stage of the frog that hatches directly from the egg.

9. **Correct answer – D**

They are called "hidden" because they exist in a microscopic world we cannot see.

10. **Correct answer – B**

Mature plants produce flowers as their reproductive organs to eventually create new seeds.

ONE OR MORE THAN ONE OPTION CORRECT TYPE

11. **Correct answer – B, C, D**

To start growing, a seed needs a combination of water, air (oxygen), and warmth (temperature).

12. **Correct answer – A, B, C**

Microorganisms are found almost everywhere on Earth, including water, food, and soil.

13. **Correct answer – A, B, C**

Bacteria, Algae, and Fungi are all microbes; frogs are multicellular animals.

14. **Correct answer – A, B, C**

Movement, growth, and reproduction are core signs of life; needing food is also a requirement.

15. **Correct answer – A, C**

Rocks and pencils are non-living; they do not breathe, grow, or reproduce.

ASSERTION AND REASON TYPE

16. **Correct answer – A**

Water is the "trigger" that softens the shell and starts the embryo's growth.

17. **Correct answer – C**

All living things do die, but non-living things do not grow (they only break into smaller pieces).

18. **Correct answer – A**

Spreading seeds ensures new plants don't compete with the parent for space.

19. **Correct answer – D**

Not all microbes are pathogens; many are helpful, as seen in food production.

MATRIX MATCHING TYPE

20. **Correct answer – A**

a- p, b- q, c- r, d- s, e- t

21. **Correct answer – C**

a- p, b- q, c- r, d- s, e- t

COMPREHENSION TYPE

Passage1

22. **Correct answer – C**

23. **Correct answer – B**

24. **Correct answer – B**

Passage2

25. **Correct answer – B**

26. **Correct answer – C**

27. **Correct answer – D**

Previous Year Questions

1. **Correct answer – A**

Statements (i) and (ii) correctly define stimuli and excretion, while breathing is part of respiration and desert plants have reduced leaves to save water.

2. **Correct answer – C**

While plants need light for photosynthesis, they do not die "instantly" without it; they would gradually weaken and die over a period of time.

3. **Correct answer – D**

Changes in the environment that trigger a reaction are called stimuli, and the inherent ability of a living being to react to them is known as responsiveness.

5 BODY MOVEMENTS

Solutions

Worksheet – 1

1. **Correct Answer (B) Bristles**
   Earthworms have tiny hair-like structures called setae (bristles) that help them grip the soil and move forward.
2. **Correct Answer (B) By releasing a slimy substance called mucus**
   A snail secretes mucus, which reduces friction and helps it glide smoothly over surfaces.
3. **Correct Answer (C) Thorax**
   In a cockroach, both legs and wings are attached to the thorax.
4. **Correct Answer (C) Exoskeleton**
   Invertebrates like snails and cockroaches have a hard outer covering called an exoskeleton, which protects their body and provides support.
5. **Correct Answer (C) Invertebrates**
   Animals that do not have a backbone are called invertebrates.
6. **Correct Answer (C) Muscular foot**
   A snail moves with the help of its muscular foot, which helps it glide slowly over surfaces.
7. **Correct Answer (A) Head, thorax, abdomen**
   The body of a cockroach is divided into three parts: head, thorax, and abdomen.
8. **Correct Answer (C) Lack a backbone**
   Invertebrates are animals that do not have a backbone.
9. **Correct Answer (C) Body structure and adaptations**
   The diversity of movement in animals depends mainly on their body structure and biological adaptations, which help them survive in different environments.
10. **Correct Answer (B) Survive and function in their environments**
    The diversity of movement in animals depends mainly on their body structure and biological adaptations, which help them survive in different environments.

WORKSHEET-2

1. **Correct Answer (B) Muscles, cartilage, ligaments, and tendons.**
   Along with bones, muscles, cartilage, ligaments, and tendons work together to help the body move smoothly and efficiently.
2. **Correct Answer (C) To reduce water resistance and allow for easy swimming.**
   A fish's body is described as streamlined because its shape helps reduce water resistance, allowing it to swim smoothly and quickly.
3. **Correct Answer (B) By using loops that press against the ground to give a push**
   A snake moves forward by forming loops in its body and pressing them against the ground to push itself ahead.
4. **Correct Answer (B) Because they are hollow and make the body light for flight**
   Bird bones are called pneumatic bones because they are hollow and filled with air spaces, which make the body lighter and help in flight.
5. **Correct Answer (B) They have turned into flipper-like paddles for swimming**
   Penguins' wings are modified into flipper-like paddles, which help them swim efficiently in water instead of flying.

6. **Correct Answer (B) Endoskeleton**
The internal framework of vertebrates is called the endoskeleton, which supports and protects the body.
7. **Correct Answer (B) Balance**
The dorsal, pelvic, and anal fins mainly help the fish maintain balance and stability while swimming.
8. **Correct Answer (B) Slithering**
The movement of a snake is called slithering.
9. **Correct Answer (B) Two legs**
Humans walk upright mainly on two legs.
10. **Correct Answer (B) Four limbs**
Humans Crawl using Four Limbs.

WORKSHEET-3

1. **Correct Answer (C) The lower jaw**

The lower jaw (mandible) is the only movable bone in the human skull.

2. **Correct Answer (C) Femur**

Femur is considered as the longest bone in the body.

3. **Correct Answer (B) Cartilage**

Cartilage is a smooth, soft tissue that covers the ends of bones at joints. It acts as a shock absorber and prevents bones from rubbing against each other.

4. **Correct Answer (C) The wrist and ankle**

Gliding joints are found in the wrist and ankle, where bones slide over one another to allow smooth movement.

5. **Correct Answer (B) Pivot joint**

The pivot joint between the first two vertebrae (atlas and axis) allows you to move your head from side to side.

6. **Correct Answer (A) 206**

Generally, Human skeleton consists of 206 bones.

7. **Correct Answer (B) Brain and Sense Organs**

The skull protects the brain and main sense organs such as the eyes, ears, and nose.

8. **Correct Answer (B) 12**

The human rib cage has 12 pairs of ribs (a total of 24 ribs).

9. **Correct Answer (B) Arms**

The pectoral girdle supports the arms and connects them to the body.

10. **Correct Answer (B) Blood Cells**

Bone marrow is responsible for the formation of blood cells (red blood cells, white blood cells, and platelets).

Competitive Corner

SINGLE OPTION CORRECT TYPE

1. **Correct Answer (B) To support powerful flight muscles**

In birds, the breastbone (sternum) is large and strong. It provides attachment for powerful flight muscles that help in flying.

2. **Correct Answer (B) Radius and ulna**
The radius and ulna are the two bones found in the human lower arm, connecting the elbow to the wrist.
3. **Correct Answer (B) Because each loop of their body presses against the ground to push them forward.**
Snakes move in a curved or zigzag pattern because each loop of their body presses against the ground, helping them push forward and move efficiently.
4. **Correct Answer (B) Appendicular skeleton**

The appendicular skeleton includes the arms, legs, and their supporting girdles (pectoral and pelvic girdles).

5. **Correct Answer (B) Soft, flexible, or covered with a hard exoskeleton**
The body of an invertebrate is usually soft and flexible, or it may be protected by a hard exoskeleton.
6. **Correct Answer (B) Worms like earthworm**
A hydrostatic skeleton is found in worms like earthworms, where fluid-filled body cavities help in movement and support.

7. **Correct Answer (A) Walking, climbing, jumping, swimming**
   Humans are capable of many movements such as walking, climbing, jumping, and swimming.
8. **Correct Answer (A) Patella**
   The knee cap is called the patella, and it protects the knee joint.
9. **Correct Answer (B) Two or more bones meet**

   A joint is the place where two or more bones meet, allowing movement and flexibility.
10. **Correct Answer (C) Synovial fluid**

    Synovial fluid is a slippery fluid found in many joints that reduces friction and allows smooth movement of bones.

ONE OR MORE THAN ONE OPTION CORRECT TYPE

11. **Correct Answer (A, B, C, D) Fixed joints, slightly movable joints, freely movable joints, Sliding joints.**
    The human body has different types of joints. Fixed joints don't move (like in the skull). Slightly movable joints move a little (like in the spine). Freely movable joints move a lot (like the shoulder and hip). Sliding joints are a type of freely movable joint where bones slide over each other (like in the wrist and ankle).
12. **Correct answers are (A, B, and D). Maintaining balance while walking and landing, helping in direction change during flight, Assisting in gliding.**
    Tail feathers (scientifically known as rectrices) are essentially the "multi-tool" of a bird's anatomy. While they might look like they are just for show, they are critical for survival and movement.
13. **Correct answers are (A, B, and C). Support and giving shape to body, Protection of vital organs, Helping in movement.**

    The human skeleton supports the body, protects organs like the brain and heart, and helps in movement by working with muscles.
14. **Correct answers are (A, B, and C). No backbone, soft body or exoskeleton, Hydrostatic skeleton in some.**

    Invertebrates lack a backbone, may have a soft body or a hard exoskeleton, and some, like earthworms, have a hydrostatic skeleton. Hollow bones are a feature of birds, not invertebrates.
15. **Correct answers are (A, D) Earthworm, Fish.**

    Animals with bilateral symmetry have two similar halves when divided along one plane. Examples include earthworms and fish.

ASSERTION AND REASON TYPE

16. **Correct Answer(A)**

    Both Assertion and Reason are true, and Reason is the correct explanation of Assertion. Fixed joints in the skull do not allow movement and their main function is to protect the brain.
17. **Correct Answer(B)**

    Both Assertion and Reason are true, but Reason is not the correct explanation of Assertion.

    The rib cage is part of the axial skeleton (Assertion is true).

    The reason - "rib cage has 12 pairs of ribs attached to vertebrae and sternum" is true, but it does not explain why it is called the axial skeleton, so it's not the correct explanation.
18. **Correct Answer(B)**

    Both Assertion and Reason are true, but Reason is not the correct explanation of Assertion.

    A fish uses pectoral fins to maintain balance (Assertion is true).

    Pectoral fins also help in changing direction and stopping suddenly (Reason is true), but that is not the reason they help in balance.

19. **Correct Answer(B)**
Both Assertion and Reason are true, but Reason is not the correct explanation of Assertion.
The human skeleton has 206 bones (Assertion is true).
Bones work with muscles, cartilage, ligaments, and tendons for movement (Reason is true), but this does not explain why the skeleton has 206 bones, so the Reason is not the correct explanation.

MATRIX MATCHING TYPE

20. **Correct Answer (A)**
a→P, b→Q, c→R, d→S
21. **Correct Answer (A)**
a→P, b→Q, c→R, d→S

COMPREHENSION TYPE

PASSAGE -1

22. **Correct Answer (B) Using wings and hollow bones**
23. **Correct Answer (B) Tail fin**
24. **Correct Answer (B) Worm with soft body**

PASSAGE-2

25. **Correct Answer (C) Fixed joint**
26. **Correct Answer (C) Synovial fluid**
27. **Correct Answer (C) Shoulder**

PREVIOUS YEAR QUESTIONS

1. **Correct Answer (B) Hinge joint**

A ball and socket joint allows movement in all directions, like the shoulder.

A hinge joint allows movement in one direction, like the elbow.

2. **Correct Answer (C) Muscular foot**

The given figure is snail; it moves with Muscular foot.

3. **Correct Answer (D) (i), (ii), (iii)**

All three features help fish swim:

1. Strong muscles – provide the force for movement.
2. Swim bladder – helps maintain buoyancy in water.
3. Paired and unpaired fins – help in steering, balancing, and propulsion.

4. **Correct Answer (C) Femur-Fits into the hip girdle.**

Femur is the thigh bone and fits into the hip (pelvic) girdle, forming the hip joint.

5. **Correct Answer (B) First seven pairs of ribs are called true ribs.**

Humans have 12 pairs of ribs.

First seven pairs are called true ribs because they attach directly to the sternum.

Ribs are attached to the backbone at the back and to the sternum (or indirectly via cartilage) at the front, not the other way around.

Ribs protect the heart and lungs, not the brain.

6 NATURES TREASURE

Solutions

Worksheet – 1

1. **Correct Answer (C) The world would be completely silent.**
   Sound needs a medium like air, water, or solids to travel. If there were no air, sound from voices, music, or any source traveling through air would not reach our ears, making the world silent.
2. **Correct Answer (A) They use sunlight to turn air and water into food.**
   Plants are called "solar-powered chefs" because of photosynthesis. Just like a chef uses a stove to cook ingredients, plants use sunlight as their energy source to transform carbon dioxide (from the air) and water (from the soil) into glucose, which is their food.
3. **Correct Answer (C) They take in "bad air" (Carbon Dioxide) and turn it into "good air" (Oxygen).**
   Trees and plants breathe in carbon dioxide and make oxygen using sunlight. This oxygen is the clean air we need to breathe. That's why forests are called Earth's oxygen factory.
4. **Correct Answer (C) Nitrogen**
   Most of the air around us is nitrogen (about 78%). Oxygen is much less (about 21%), and other gases like carbon dioxide and water vapor are only tiny amounts.
5. **Correct Answer (C) Water**
   Water can be:
   Solid → ice
   Liquid → water
   Gas → water vapor/steam
6. **Correct Answer (B) Collecting and storing rainwater**
   We save rainwater in tanks or ponds to use later for drinking, farming, or gardening.
7. **Correct Answer (B) Is the main source of energy**
   Sunlight gives energy for plants to make food and powers almost everything on Earth.
8. **Correct Answer (A) Absorb rainwater like a sponge**
   Trees soak up rainwater with their roots, so less water floods rivers and land.
9. **Correct Answer (B) Because about 96.5% of it is salty ocean water**
   Saltwater cannot be drunk or used for farming without special treatment.
10. **Correct Answer (B) It burns up many small meteors before they can hit the ground.**
    The atmosphere stops most meteors from hitting us by burning them up in the sky.

WORKSHEET-2

1. **Correct Answer (C) Soil.**
   The very top layer of the Earth that supports life is called soil.
2. **Correct Answer (C) Natural substances found in rocks**
   Minerals are the building blocks of rocks; they are inorganic and occur naturally.
3. **Correct Answer (A) Coal**
   Coal is the primary fossil fuel burned in power plants to generate electricity.
4. **Correct Answer (B) They take millions of years to form**
   Because they take so long to develop from ancient remains, we cannot replace them once they are used up.
5. **Correct Answer (C) Air pollution**
   Burning these fuels releases smoke, soot, and greenhouse gases into the atmosphere.

6. **Correct Answer (D) Graphite**

   The "lead" in your pencil is actually a soft mineral called graphite.

7. **Correct Answer (C) Soil**
   Just like frosting is the thin top layer on a cake, soil is the thin, vital layer covering the Earth's rocky surface.
8. **Correct Answer (B) Fossils**
   Fossils are the preserved remains or traces of plants and animals from the deep past.
9. **Correct Answer (A) To prevent erosion**
   Tree roots act like an anchor, holding the soil in place so it doesn't get washed away by rain or blown away by wind.
10. **Correct Answer (C) They release smoke and gases that cause global warming**
    The carbon dioxide released from burning these fuels traps heat in our atmosphere.

WORKSHEET-3

1. **Correct Answer (C) Water cycle**
   The Water cycle continuously recycles water through evaporation,condensation, and precipitation, making it a Renewable resource.
2. **Correct Answer (C) Oil**
   Oil is a fossil fuel that takes millions of years to form and cannot be replaced once used. Wind, water, and sunlight are constantly available.
3. **Correct Answer (C) They take millions of years to form**
   Metals and minerals are formed by geological processes over extremely long periods, so they are considered non-renewable on a human timescale.
4. **Correct Answer (B) Savings account that can become empty**
   This analogy illustrates that non renewable resources have a fixed "balance,"and every time we use some, there is less left for the future.
5. **Correct Answer (C) Sunlight**
   Solar energy is a perpetual resource that does not get "used up" by humans, unlike coal, gold, or oil.
6. **Correct Answer (C) A cookie jar that slowly becomes empty**

   This is a classic analogy for non-renewable resources, showing that while they are available now, the supply is finite and decreasing
7. **Correct Answer (B) Because it gives us light and heat and will never run out in our lifetime**

   The Sun provides an essentially infinite supply of energy for all practical human purposes, similar to a fountain that never stops flowing.
8. **Correct Answer (C) Wind**

   Wind turbines capture the kinetic energy of moving air and convert it into electricity.
9. **Correct Answer (C) gold**

   Unlike wind or sunlight, gold is a physical material extracted from the Earth; there is only a certain amount of it available to mine.
10. **Correct Answer (B) Non-renewable**

    For a resource to be renewable, it must replenish at least as fast as it is consumed. If we cut trees faster than they grow, the forest disappears, making it non-renewable in that context.

COMPETITIVE CORNER

SINGLE OPTION CORRECT TYPE

1. **Correct Answer (B) It surrounds and protects the Earth**

   Just as a chocolate shell covers a treat, the atmosphere acts as a protective layer that surrounds the entire planet, shielding us from space radiation and small meteors
2. **Correct Answer(C) It does not get used up when we use sunlight**

   The Sun is a renewable resource; its energy output is so massive that it won't run out for billions of years, regardless of how much solar power we collect

3. **Correct Answer (B) Ancient plants and animals**

   Fossils are the remains or impressions of prehistoric organisms preserved in sedimentary rocks, giving us a "window" into Earth's biological history

4. **Correct Answer(D) They store energy originally captured from the Sun millions of years ago**

   Fossil fuels (coal, oil, gas) formed from ancient plants and plankton that used photosynthesis to store solar energy. When we burn them, we are releasing that "ancient" stored sunlight

5. **Correct Answer (A) Iron ore**

   Iron ore is a non-renewable mineral. Unlike rain, sunlight, or wind (which are naturally replenished), once we mine and use all the iron ore in the ground, it is gone forever

6. **Correct Answer(B) Wind**

   Wind is simply the movement of air within the atmosphere caused by temperature and pressure differences. Without the atmosphere, there would be no air to move

7. **Correct Answer(B) In the salty oceans**

   While fresh water is scarce (around 2.5–3%), the vast majority (about 97%) of Earth's total water volume is salt water found in the oceans

8. **Correct Answer(D) The Atmosphere**

   The air becomes "thinner" (less dense) as you go higher because you are traveling through the upper layers of the atmosphere, the gaseous envelope surrounding Earth

9. **Correct Answer (B) It helps the forest act as a natural cooler for the world**

   This process is called transpiration. As trees release water vapor, it absorbs heat from the air to evaporate, which effectively "cools" the surrounding environment—much like how human sweat cools our skin

10. **Correct Answer (A) The Land**

    This uses a metaphor for the three spheres: the atmosphere is the "ocean of air," the hydrosphere is the "liquid gold" (water), and the lithosphere (land/soil) provides the solid foundation we walk on

ONE OR MORE THAN ONE ANSWER CORRECT TYPE

11. **Correct Answer (D) Only 1% of Earth's total water is fresh and easily available.**

    If a bucket represents all Earth's water, 97% is salt water (unusable) and 2% is frozen in ice caps (hard to reach). That leaves just **1%** in our rivers, lakes, and shallow underground wells. This tiny "teaspoon" is all we have for 8 billion people

12. **Correct Answer (A,B,C,D)**

    A (Oxygen): They breathe out the oxygen we need.
    B (Sponge): Their roots and leafy floors soak up rain, preventing sudden "flash floods."
    C (Erosion): Roots act like biological glue, holding the soil so it doesn't wash away in a storm.
    D (Cooling): Through "transpiration," trees release moisture that cools the air, acting like a giant, natural AC.

13. **Correct Answer (A,C) A) Sunlight and C) Wind**

    Renewable means “it comes back.” No matter how many solar panels or wind turbines we build, the Sun will still shine and the wind will still blow tomorrow. Coal and Gold are finite; once they are pulled from the ground, they don’t grow back.

14. **Correct Answer (A,B,D) A) Iron, B) Coal, and D) Gold**

    These are Non-Renewable. They were formed millions of years ago. Using them is like spending money from a savings account where no one is adding more cash. Eventually, the "account" (the Earth's crust) will hit zero.

15. **Correct Answer (A,B,D)**

    A (Nutrients): Soil is a "pantry" full of minerals that plants need to grow.

B (Anchor): It provides the physical grip for roots so trees don't fall over.
D (Filter/Sponge): It absorbs rainwater, filtering it as it moves down to refill our underground water supplies.

ASSERTION AND REASONING TYPE

16. **CORRECT ANSWER (A) Both Assertion and Reason are true, and Reason is the correct explanation of Assertion.**
The Assertion is true because wind is a renewable/flow resource.
The Reason explains why it never runs out: wind is powered by the Sun, which constantly heats the Earth unevenly, keeping the air in motion. Since the Sun won't stop shining, the wind won't stop blowing.
17. **CORRECT ANSWER (A) Both Assertion and Reason are true, and Reason is the correct explanation of Assertion.**
Even though we can recycle a gold ring or an iron bar, the actual "raw material" (ore) inside the Earth is finite. Because it takes millions of years to form, we cannot "grow" more iron once the mines are empty. This makes them depletable.
18. **CORRECT ANSWER (D) Assertion is false, but Reason is true.**
The Assertion is false because gold and iron are non-renewable, not renewable.
The Reason is true—they do take millions of years to form—but that is actually the reason why they are not renewable.
19. **CORRECT ANSWER (A) Both Assertion and Reason are true, and Reason is the correct explanation of Assertion.**
Air is essential for life (Assertion) specifically because of the Oxygen it contains. Our cells use that oxygen in a process called respiration to break down nutrients from our food and turn them into the energy we need to move and grow.

MATRIX MATCH TYPE

20. **CORRECT OPTION (A) a→P, b→Q, c→R, d→S**
21. **CORRECT OPTION (A) a→P, b→Q, c→R, d→S**

COMPREHENSION TYPE

Passage-1

22. **CORRECT OPTION (C) They take thousands of years to form**
23. **CORRECT OPTION (C) Minerals**
24. **CORRECT OPTION (B) Gold**

Passage-2

25. **Correct option (B) Petrol and diesel**
26. **Correct option (C) Coal**
27. **Correct option(B) Ancient Plants and Animals**

Previous year questions

1. **Correct Answer (A) T F T F**
   i. True: Air is matter because it has mass and occupies space.
   ii. False: Dust particles are always present in the atmosphere (smoke, pollen, etc.).
   iii. True: We inhale about 21% oxygen and exhale about 16% (our bodies use some of it).
   iv. False: Nitrogen is used in chip packets, not Carbon Dioxide, because Nitrogen is unreactive and keeps chips crunchy.
2. **Correct Answer (D) Air is incompressible**
This is incorrect because air is a gas, and gases are highly compressible. You can see this when you squeeze a balloon or use a bicycle pump to force a lot of air into a small tire.
3. **Correct Answer (A) F T T F**
   i. False: Water seeping into the ground is called Infiltration; precipitation is rain, snow, or hail falling from the sky.
   ii. True: As you go higher (higher altitude), the air becomes "thinner" or less dense.
   iii. True: Water helps disperse seeds (like coconuts) and some aquatic plants rely on it for pollination.
   iv. False: The Mesosphere is generally the coldest layer; the Stratosphere actually warms up slightly as you go higher due to the ozone layer.

7 Habitat and Adaptation

Solutions

Worksheet – 1

1. **Correct answer – B**

   The statement defines the environment as the sum of all surrounding biotic and abiotic factors that impact an organism's life.

2. **Correct answer – C**

   Abiotic components are the non-living parts of an ecosystem, such as sunlight, soil, and temperature.

3. **Correct answer – B**

   The statement states that biotic components are classified into three groups (producers, consumers, decomposers) based on how they get nutrients.

4. **Correct answer – C**

   Producers use sunlight, chlorophyll, CO_2, and water to create their own food internally.

5. **Correct answer – A**

   Decomposers break down dead organic matter, which recycles essential nutrients back into the environment for plant use.

6. **Correct answer – D**
   Omnivores are defined in the text as organisms, like bears and humans, that consume both plant and animal products.

7. **Correct answer – B**
   The statement explicitly identifies secondary consumers as exclusively flesh-eating animals, such as lions and wolves.

8. **Correct answer – A**

   While animals get food from plants, plants rely on animals to move pollen and spread seeds for reproduction.

9. **Correct answer – C**
   A habitat is described as a "natural home" that provides every necessity for an organism to survive and thrive.

10. **Correct answer – D**
    The text lists sandy deserts, along with grasslands and rainforests, as examples of unique habitats on Earth.

worksheet – 2

1. **Correct answer – B**

   Adaptation is a genetic change over generations, whereas acclimatization is a temporary adjustment by an individual to a new environment.

2. **Correct answer – A**

   The fleshy tissue acts as a reservoir, and the waxy coating prevents precious moisture from evaporating in the heat.

3. **Correct answer – C**

   By staying underground or in shade during the day, these animals avoid the desert's extreme daytime temperatures.

4. **Correct answer – D**
   The sloping branches and conical structure allow heavy snow to slide off, preventing the limbs from breaking.

5. **Correct answer – B**
   High-altitude air is thin, so larger organs are necessary to efficiently capture and distribute limited oxygen to the body.

6. **Correct answer – C**
   In open spaces with little cover, detecting predators early and being able to outrun them is essential for survival.

7. **Correct answer – A**

   Contrary to the myth about water, the hump is a concentrated energy reserve that the camel metabolizes when food isn't available.

8. **Correct answer – B**

These animals have hard outer edges and soft, grippy pads on their hooves that act like natural climbing shoes.

9. **Correct answer – C**

Grass is difficult to break down, so these animals need strong, flat teeth to grind the cellulose into a digestible pulp.

10. **Correct answer – D**

The fur provides critical insulation against the cold while helping the bear blend into the snow to sneak up on prey.

worksheet 3

1. **Correct answer – B**

Because rainforest soil is thin, trees have shallow roots and need massive buttress roots for stability and support.

2. **Correct answer – C**

The waxy surface and pointed "drip tips" prevent water from accumulating on the leaves in the wet environment

3. **Correct answer – A**

The trees grow so close together that their branches spread out to form a thick layer that blocks nearly all sunlight from the floor.

4. **Correct answer – B**

Animals living in the topmost layer stay in the trees and rely on water collected in natural tree cavities.

5. **Correct answer – D**

By definition, freshwater habitats like rivers and ponds are characterized by having very little salt compared to the ocean.

6. **Correct answer – C**

These air-filled channels act like "floats" to keep the plant upright or on the surface while helping with gas exchange.

7. **Correct answer – B**

Hydrilla is specifically described as being rooted at the bottom while staying completely underwater with flexible, ribbon-like leaves.

8. **Correct answer – C**

Unlike fish with gills, these mammals must surface to breathe air through their dorsal blowholes.

9. **Correct answer – D**

These specific animals are adapted to spend their entire lives feeding and traveling within the high canopy layer.

10. **Correct answer – A**

The primary distinction is salinity; oceans (marine) have high salt levels, while freshwater bodies have low concentrations.

Competitive Corner

SINGLE OPTION CORRECT TYPE

1. **Correct answer – C**

Deserts are characterized by very low rainfall, requiring plants and animals to have special adaptations to conserve water.

2. **Correct answer – B**

Acacia trees are well-adapted to arid regions with deep roots to reach underground water and small leaves to minimize water loss.

3. **Correct answer – A**

Forests cover approximately 31% (roughly one-third) of the Earth's total land surface.

4. **Correct answer – D**

The "Batta meka pitta" (Great Indian Bustard) is a critically endangered bird found in the Rollapadu Wildlife Sanctuary in the Kurnool district.

5. **Correct answer – B**

Also known as water lettuce, this plant is a free-floating hydrophyte that stays entirely on the water's surface.

6. **Correct answer – A**

Cacti are classic desert plants that use succulent stems to store water and spines to protect themselves and reduce transpiration.

7. Correct answer – D

Migration allows birds to move to areas with better climate and more food, which are essential for successful breeding.

8. Correct answer – C

The wide variety of habitats on Earth supports a massive range of different species, representing the diversity of life.

9. Correct answer – B

Organisms like green plants are called producers because they synthesize their own food using sunlight through photosynthesis.

10. Correct answer – A

While its large leaves and flowers float on the surface, the lotus is rooted in the soil at the bottom of the water body.

ONE OR MORE THAN ONE OPTION CORRECT TYPE

11. Correct answer – A, D

Biotic components refer to all living beings in an environment, which include both birds and animals.

12. Correct answer – A, B, C

These are the abiotic components that make up the physical environment of a habitat; fuel is not a natural component.

13. Correct answer – A, B

Both rivers (freshwater) and lagoons (saltwater/brackish) are water-based environments, whereas grasslands and forests are terrestrial.

14. Correct answer – A, C, D

A habitat must provide the essential physical (air/water) and structural (shelter) needs for an organism to survive.

15. Correct answer – B, C, D

Animals migrate or move to satisfy basic survival needs like nourishment, safety, and the continuation of their species.

ASSERTION AND REASON TYPE

16. Correct answer – A

Bacteria and fungi (decomposers) are vital for the ecosystem. These microorganisms break down complex organic compounds from dead plants and animals into simpler nutrients (like nitrogen, phosphorus, and potassium).

17. Correct answer – C

The Assertion is true, but the Reason is false because oxygen is actually less (scarce) at high altitudes, which is why the yak needs a bigger heart/lungs.

18. Correct answer – D

The Assertion is false because the scenario describes Acclimatisation, not adaptation. The Reason correctly defines Acclimatisation.

19. Correct answer – D

Hydrilla is a submerged aquatic plant. It does not have thick, broad leaves; instead, it has thin, narrow, and often ribbon-like leaves. The thin, ribbon-like structure is a specific adaptation for submerged plants.

MATRIX MATCHING TYPE

20. Correct answer – C

a-Q, b-R, c-S, d-T, e-P

21. Correct answer – D

a-R, b-S, c-P, d-Q

COMPREHENSION TYPE

Passage1

22. Correct answer – C

23. Correct answer – B

24. Correct answer – A

Passage2

25. Correct answer - B

26. Correct answer - C

27. Correct answer – A

Previous Year Questions

1. **Correct answer – B**

   Group II represents polar (cold) animals where Rattlesnake (desert) is the misfit, and Group III represents desert animals where Red-eyed tree frog (rainforest) is the misfit; swapping Rattlesnake with Camel fixes the desert category.

2. **Correct answer – B**

   Unscrambling "MNOVIROES" gives OMNIVORES, which are animals that eat both plants and the flesh of other animals.

3. **Correct answer – C**

   The table (though partially rendered) implies Z has specific vertebrate features like lungs or feathers/flippers, which an earthworm (an invertebrate that breathes through its skin) does not possess.

4. **Correct answer – D**

   Cow and Goat are Herbivores (P); Lion, Tiger, Spider, and Lizard are Carnivores (Q); while Crow, Human, and Bear are Omnivores (R).

5. **Correct answer – D**

   This statement is incorrect because plants store food in various parts (roots, stems, leaves, seeds), not only in fruits.

6. **Correct answer – B**

   Large water intake, nostril flaps, and long eyelashes are desert adaptations (Camel), and aestivation is for hot/dry periods; Polar bears need thick fur and padded feet for cold/ice.

7. **Correct answer – C**

   Group III consists of polar/tundra animals, but the Gila monster is a desert reptile; the snowshoe hare is a proper fit for a cold climate.